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On the Prey Size Preferences of Cod and Dab

By
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ABSTRACT

The prey size preference of predators is described as a log-normal frequency distribution of predator-weight/prey-weight except when some prey species is exceptionally available to the predator. On estimating the parameters of the distribution from stomach contents differences in abundance of prey of each size in the environment must be accounted for. For bottom animals numbers in the environment were found inversely proportional to body weight.

INTRODUCTION

On estimating the adequacy of one animal as prey to another the size relationship of the two is an obvious subject for investigation. Few authors, apparently, have been concerned with this problem with respect to fishes as predators. Even a cursory exposition may therefore be of some interest.

There are two major problems to be investigated namely, the size frequency of food animals in the predator stomach and the availability of these animals in the environment. In a simple approach the latter problem may be reduced to investigating the frequency of potential prey as a function of its size.

A solution to these two problems should disclose the prey size preference of the predator. There may be difficulties in the limits, however, because several unsuitably small prey animals close together may be perceived by the predator as one larger food item and picked in one mouthful. Also the habit of some predators, like the dogfish, to chop off suitable mouthfuls from the sides of other fishes too large to be eaten whole, are likely to confuse the issue.

In the following, the problem of size frequency in the environment is dealt with first because its solution is prerequisite to the treatment of data on stomach contents. Suitable data is available only for bottom sampler material. This is not serious because the results shall be used on predators mainly eating bottom invertebrates.

THE SIZE FREQUENCY OF BOTTOM ANIMALS

As a starting point assume that numbers per unit area is a simple function of body weight, independent of the size range in the material so that it is as applicable to micro bottom samples as to trawl data. A realistic expression may then be: –

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$$N_w = N_o w^n \qquad (1)$$

where N: numbers per unit area and w: body weight; n is the important parameter.

The obvious and crude way to check equation 1 and estimate n is by linearizing: –

$$\log N_w = \log N_o + n \log w \qquad (2)$$

and then use the method of least squares. Because small animals are more numerous than big ones, n obviously is a negative quantity. Moreover, $n \leq -1$ may be expected because $n > -1$ implies that the biomass increases with body size which is not feasible as a general hypothesis.

In any one data set there will probably be a defect in numbers of the smallest animals because mesh selection or a similar agent has been at work.

Results of general interest should be based preferably on extensive materials in which each animal has been weighed. No such material is available, but the listings of bottom sampler contents by e.g. Petersen (1914) and Blegvad (1930) provide reasonable approximations. For samples covering one or a few square metres these authors state the weight and numbers per sq. m. of each of many species.

Calculating the mean size of each species in each list, adopting log scale weight groups and summing the numbers in each group for the entire material produces data against which the hypothesis can be tested. Yet, there are awquard points to be considered. Both authors weigh with an accuracy of .1g and allot a weight of .1g to even the tiniest animal when it is the only specimen of its kind in the sample. Thus, mean weights of less than .1g, approximately, are unreliable.

Petersen occasionally indicates large numbers of very small animals by an infinity sign. Also, he sometimes writes "fragments" in stead of a number. This is probably less disastrous to the result. Observations of "infinity" and "fragments" were excluded from the investigation. So were the weights of some not readily counted species (e.g., sponges, bryozoans, hydroids).

In both papers the numbers and weights per sq. m. listed have been used directly, without weighting by the actual size of each sample.

Petersen's Data.

2963 observations of weight-and-numbers were used. Numbers were grouped into weight classes with a class interval of .25 on a $\log_e$ scale (i.e., the class midpoint of any weight class is 28.4 pCt. bigger than the preceding one). See Table 1 and Fig. 1 A. For reasons related above body weights of less than .1g were omitted from the analysis. In the upper size range zero observations, and the observations following them, were rejected in order to satisfy the log transformation.
There remained 30 observations for the estimation of n and its variance:–

n	−1.009
Variance of n	.01397
95 % conf. intervals of n	−1.251 to −.767
Variance about regr. line	1.962

The hypothesis of linearity cannot be tested with such data but seems acceptable on inspection (Fig. 1 A), considering the poor quality of the data.

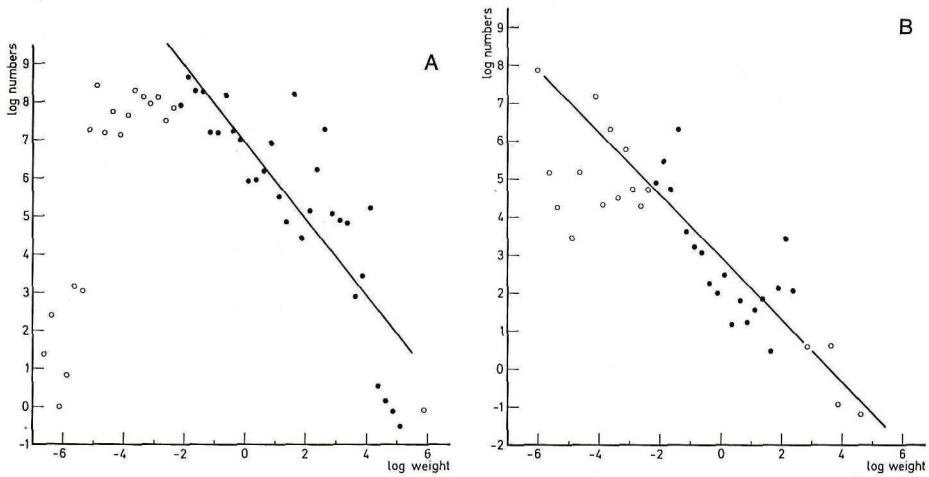


Fig. 1. Size frequency of bottom animals in Danish waters. Black dots indicate the more reliable points (cf. text). A: From Petersen 1914. B: From Blegvad 1930.

Blegvad's Data.

The material is much smaller, 495 observations of weight-and-numbers. It was treated like Petersen's, see Table 1 and Fig. 1B. Using only the size range within which there are no zero observations and omitting weights <.1g leaves 19 observations for parameter estimation:–

n	−.824
Variance of n	.03382
95% conf. intervals of n	−1.212 to −.436
Variance about regr. line	1.205

Small animals are better represented than in Petersen's material and the linearity is more convincing (cf. Fig. 1B). Likely enough the technique has been refined in the years between the two investigations.

Conclusion.

In both cases, $n \approx -1$. Until further information becomes available the relationship

$$N_w = N_0 w^{-1} \tag{3}$$

is as good as any and has the virtue of being simple.

THE PREDATOR/PREY SIZE RELATIONSHIP

Size is described by body weight throughout because body weight is a simple statistic to measure. The maximum width or girth of the prey might be a more relevant measure of its suitability as food for the predator in question. Such data is not available except in Okada and Taniguchi (1971) who unfortunately measured only the biggest food item in each stomach.

Two sets of observations are available. One set, on dab, was used to develop an empirical predator-size/prey-size model. The other set, on cod, was used for testing the model.

Table 1. Size frequency of animals in bottom sampler material.

Body weight (class midpoints)		Numbers			
		Petersen 1914		Blegvad 1930	
w, g.	$\log_e$ w	N	$\log_e$ N	N	$\log_e$ N
.00132	–6.625	4.00	1.386		
.00170	–6.375	11.00	2.398		
.00219	–6.125	1.00	0		
.00281	–5.875	2.00	.693		
.00361	–5.625	24.00	3.178	177.50	5.179
.00463	–5.375	21.10	3.049	69.50	4.241
.00595	–5.125	1432.70	7.267	0	
.00764	–4.875	4703.80	8.456	30.80	3.428
.00980	–4.625	1345.20	7.204	181.10	5.199
.0126	–4.375	2307.05	7.744	0	
.0162	–4.125	1253.35	7.134	1322.30	7.187
.0208	–3.875	2111.08	7.655	74.90	4.316
.0266	–3.625	4082.90	8.315	554.00	6.317
.0342	–3.375	3459.30	8.149	90.90	4.510
.0439	–3.125	2966.85	7.995	322.40	5.776
.0564	–2.875	3422.20	8.138	117.00	4.762
.0724	–2.625	1838.82	7.517	73.30	4.295
.0930	–2.375	2556.37	7.846	112.80	4.726
.119	–2.125	2807.09	7.940	134.80	4.904
.153	–1.875	5803.97	8.666	239.90	5.480
.197	–1.625	4031.45	8.302	113.50	4.732
.253	–1.375	3969.16	8.286	547.50	6.304
.325	–1.125	1347.37	7.206	36.40	3.595
.417	– .875	1353.13	7.210	25.30	3.231
.535	– .625	3660.60	8.205	21.30	3.059
.687	– .375	1434.77	7.269	9.60	2.262
.882	– .125	1115.83	7.018	7.40	2.001
1.13	.125	378.55	5.936	11.90	2.477
1.45	.375	393.75	5.976	3.30	1.194
1.87	.625	500.51	6.216	6.20	1.825
2.40	.875	1052.00	6.958	3.50	1.253
3.08	1.125	250.40	5.523	4.80	1.569
3.96	1.375	128.10	4.853	6.40	1.856
5.08	1.625	3747.37	8.229	1.60	.470
6.52	1.875	85.20	4.445	8.50	2.140
8.37	2.125	172.40	5.150	31.30	3.444
10.8	2.375	516.38	6.247	7.90	2.067
13.8	2.625	1522.60	7.328	0	
17.7	2.875	160.10	5.076	1.80	.588
22.8	3.125	134.80	4.904	0	
29.2	3.375	126.03	4.837	0	
37.5	3.625	18.90	2.939	1.80	.588
48.2	3.875	31.60	3.453	.40	.916
61.9	4.125	192.00	5.257	0	
79.4	4.375	1.80	.588	0	
102	4.625	1.20	.182	.30	1.204
131	4.875	.90	–.105		
168	5.125	.60	–.511		
216	5.375	0			
277	5.625	0			
356	5.875	.90	–.105		

Dab Stomachs

Arntz (1971) published the food contents of 1736 dab stomachs from the Kieler Bay in the Western Baltic. They were collected at all seasons.

The observations cannot be referred to size group of dabs, but an approximate mean weight of 125g can be estimated. The majority were about 25 cm long.

For each species of food animals is stated the numbers and weight in the whole material by quarters of the year. One "species" is "Fischerei-abfälle" whose abundance on the feeding grounds is elusive. This category was omitted.

The number of food items was 40371 with a total weight of 1422.68g. The overall mean weight of animals consumed was .0352g. There are 98 observations of numbers-and-weight of kinds of food animals. Individual mean weights were calculated and the numbers grouped with class intervals of .2g, see Table 2A.

Table 2A. Dab food size (from Arntz). Observed distribution.

Mean weight, g. (class midpoints)	Numbers	Log mean weight (class midpoints)	Numbers	Cumulative frequencies	Fractiles u_p
.1	39378	−6.5	214	.0053	−2.56
.3	767	−5.5	185	.0099	−2.33
.5	18	−4.5	23596	.59	.24
.7	39	−3.5	12332	.900	1.28
.9	39	−2.5	3027	.975	1.96
1.1	0	−1.5	790	.9944	2.53
1.3	0	−0.5	97	.9968	2.72
1.5	0	+0.5	130	1	4.21
1.7	119				
1.9	11				

Table 2B. Dab food size (from Arntz). Simulated environment. (Numbers weighted by the mean weight, cf. text)

Mean weight, g. (class midpoints)	Total weight	Log mean weight (class midpoints)	Total weight	Cumulative frequencies	Fractiles u_p
.1	900.63	−6.5	.44	.00031	−3.43
.3	225.24	−5.5	.61	.00073	−3.18
.5	8.94	−4.5	290.46	.20	− .82
.7	28.67	−3.5	379.09	.47	− .07
.9	35.92	−2.5	225.86	.63	.33
1.1	0	−1.5	229.01	.79	.81
1.3	0	−0.5	73.93	.84	1.01
1.5	0	+0.5	223.29	1	4.50
1.7	203.35				
1.9	19.94				

The distribution is markedly skew. A log transformation, however, makes the distribution almost symmetrical (Table 2A again). The cumulative frequency in the logarithmic transformation was turned into fractiles. The linear relationship

$$u_p = m + \frac{1}{s} \log w \qquad (4)$$

where u_p: fractile
m : logarithmic mean
s : standard deviation

is realized if the distribution is log-normal. Inspection of Fig. 2A shows that eq. 4 is a fair description when outlying observations are omitted. So far, however, was disregarded the information acquired in the preceding paragraph on the availability of bottom animals as a function of their size (weight). Numbers were found to be approximately inversely proportional to body weight (eq. 3).

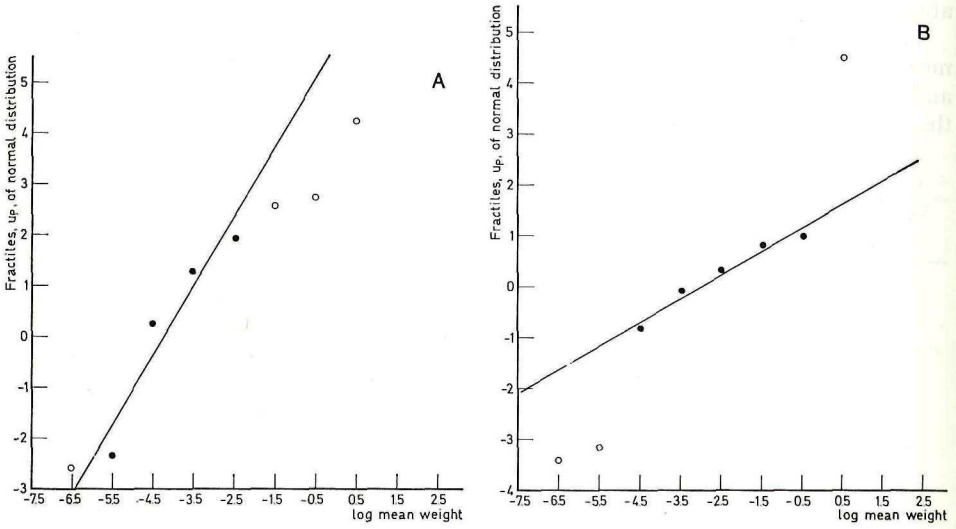


Fig. 2. Dab food size (from Arntz, cf. Table 2). Fractiles of a log-normal distribution. Outlying observations (open circles) omitted from regression analysis. A: Observed distribution. B: Simulated environment, all sizes of prey equally abundant.

To simulate an environment with all sizes of food animals equally abundant the numbers in each weight group could be multiplied by their weight. In the present case that implies counting, not numbers from Arntz's Table 2, but weights.
An illustration:–

Harmothoe sarsi, 1st quarter,

Numbers, N	307
Total weight, W	30.25g
Mean weight, $\overline{w}$	.098534g
N weighted by $\overline{w}$	30.25g = W

The overall mean size of items consumed in the simulated environment is $\Sigma(W \times \overline{w})/\Sigma W = .383$g as against .0352g in the actual environment.

The size frequency distribution of the prey in the simulated environment was investigated as in the actual environment, cf. Table 2B and Fig. 2B. Disregarding outlying observations the linearity is acceptable on inspection.

Formalized, the prey size model presented above has the shape of

$$p(\log w) = \frac{1}{s\sqrt{2\pi}} \exp\left(-\frac{(\log w - \overline{\log w})^2}{2\,s^2}\right)$$

where w is the size of prey and predator size remains constant.

Generalizing to any size of predator the variable log w is replaced by $\log(w_i/w_j)$ where w_i is the weight of the predator, w_j the weight of the prey. We then have

$$p\left(\log \frac{w_i}{w_j}\right) = \frac{1}{s\sqrt{2\pi}} \exp\left(-\frac{\left(\log \frac{w_i}{w_j} - \overline{\log \frac{w_i}{w_j}}\right)^2}{2\,s^2}\right). \tag{5}$$

Simulating an environment with equal numbers of food items of all sizes must be done for the possible prey of each predator. Weighting of numbers is therefore made by the factor w_j/w_i:–

$$N_j' = N_j \cdot \frac{w_j}{w_i} = \frac{W_j}{w_i}. \tag{6}$$

This reduces to $N_j'' = N_j w_j = W_j$ for w_i = constant, as it should.

When applied to stomach contents data directly eq. 5 describes the size frequency of consumed prey without informing on the prey size preference of the predator.

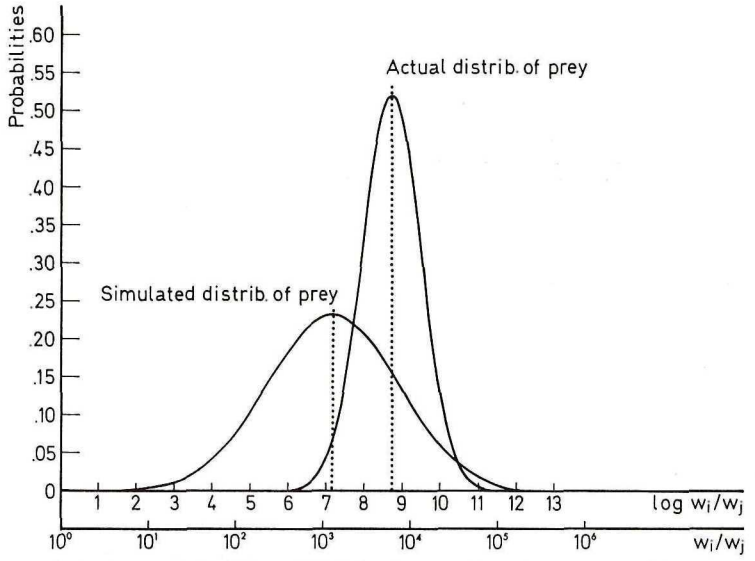


Fig. 3. Prey size selection by dabs (Arntz 1971) in the actual environment and in a simulated one in which all sizes of food items are equally abundant. w_i: weight of predator, w_j: weight of prey.

After transformation of the frequencies by such means as eq. 6 the issue is different. Eq. 5 then describes the prey size preference of the predator. An application to both the actual and the simulated environment of the dabs is illustrated in Fig. 3. The estimation is based on data from Table 2 except that a log-weight group interval of

.25 was adopted, in stead of 1. Outlying observations were included. Predator size (125g, constant) is the approximate mean size of the dabs investigated. It should be remembered that the weight of individual dabs varied from about 50g to about 300g. The variance is overestimated because the smaller dabs very likely ate the smaller prey.

The parameter estimates are:–

	$\text{Mean}\left(\overline{\log \frac{w_i}{w_j}}\right)$	Variance
Actual environment	8.723	.5806
Simulated environment	7.161	2.978

The median in the predator/prey size relationship is found by antilogging the means:–

Actual environment	6100:1
Simulated environment	1300:1

The accuracy of these figures is elusive because the numbers of degrees of freedom is problematic. Estimating from the variances about the regression lines in Fig. 2 presents difficulties because the observations are not stochastically independent and have different variances.

The next step in the analysis is testing the empirical model (eq. 5) against a different material. There are two points to be investigated: the log-normality of the frequency distribution, and the independence of that distribution of predator size.

Cod Stomachs

Dr. O. Bagge most kindly placed at disposal an unpublished material of data on cod stomach contents: –

a. from round-the-clock hauls in the Sound between Denmark and Sweden, in Sept. 1965
b. from two supplementary hauls in the Sound, about noon in Sept. and Nov. 1965
c. from a cutter landing from the Western Baltic, in Nov. 1967; these records are not referred to hour of day.

The data state, for each cod or for length groups of cod, the numbers and total weight of each species of prey (mostly bottom animals). Cod lengths were converted to weights by assuming $w = .01 \cdot \text{length}^3$.

The condition of the stomach contents is stated, and some material (particularly from day hauls) could not be identified. Only records on identifiable prey in good condition of preservation were accepted. This made a total of 542 useful records, 147 of which from day hauls (from noon to 3 p.m., actually).

Most of the cod investigated weighed 200–500g, the full size range being 10–2800g. Table 3 summarizes the predator/prey size relationship expressed as log (w_i/w_j) in the actual environment (i.e., number of occurrences) and in a simulated environment in which each cod has access to equal numbers of all size groups of food items assuming, that eq. 3 describes the size frequency of prey in the actual environment. Records of cumaceans are kept separate for reasons explained below.

Fig. 4A is a plot of the cumulative distribution of log (w_i/w_j) in the simulated environment. The curve is conspicuously non-linear, but suggests a heterogeneity dissectable into two log-normal distributions as indicated by the regression lines.

Table 3. Cod food size: the predator/prey relationship expressed on a log scale. w_i: cod weight; w_j: prey weight.

log (w_i/w_j) (class midpoints)	Numbers of prey animals			
	Actual environment (N_j)		Simulated environment ($N'_j = N_j w_j / w_i$)	
	Cumaceans omitted	Cumaceans	Cumaceans omitted	Cumaceans
3.375	5		.1673	
3.625	6		.1488	
3.875	8		.1692	
4.125	12		.1991	
4.375	44		.5607	
4.625	45		.4356	
4.875	64		.4903	
5.125	55		.3393	
5.375	72		.3370	
5.625	71		.2428	
5.875	85		.2598	
6.125	23	7	.0505	.0137
6.375	102		.1643	
6.625	192		.2636	
6.875	57	14	.0610	.0161
7.125	79		.0650	
7.375	72		.0454	
7.625	38	10	.0184	.0053
7.875	44	9	.0177	.0039
8.125	33	2	.0101	.0005
8.375	16	123	.0037	.0288
8.625	17	114	.0030	.0198
8.875	14	592	.0020	.0803
9.125	53	808	.0064	.0865
9.375	108	4901	.0101	.3940
9.625	11	1588	.0007	.1146
9.875	36	1110	.0018	.0599
10.125	38	6747	.0016	.2718
10.375	39	546	.0012	.0160
10.625	2	3111	.0000	.0809
10.875	16		.0003	
11.125		1807		.0264
11.375	5		.0006	
11.625	3		.0000	
11.875				
12.125	1		.0000	
12.375				
12.625				
12.875				
13.125				
13.375		27		.0000

There is in the material a remarkable abundance of cumaceans weighing about .01g each and often occurring by hundreds at a time in a stomach. Among the 542 records 145 are of cumaceans. Considering day-time hauls only we find 147 records, 3 of which are records of cumaceans.

Splitting the material into records of cumaceans and records of other food items produces Fig. 4B. The agreement of each data set with a log-normal distribution is

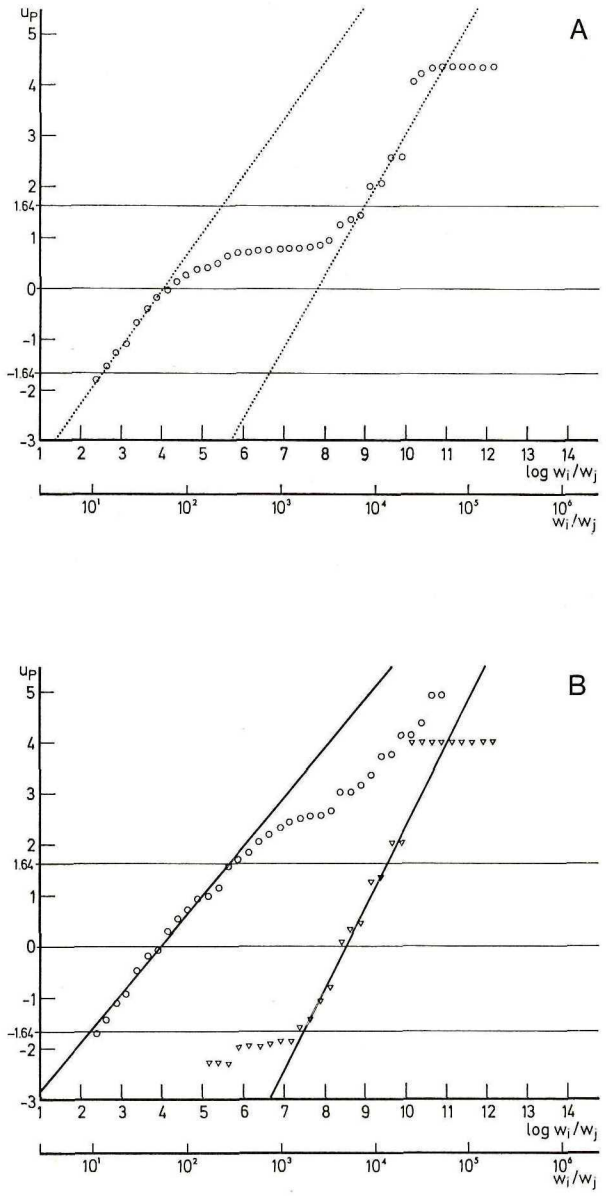
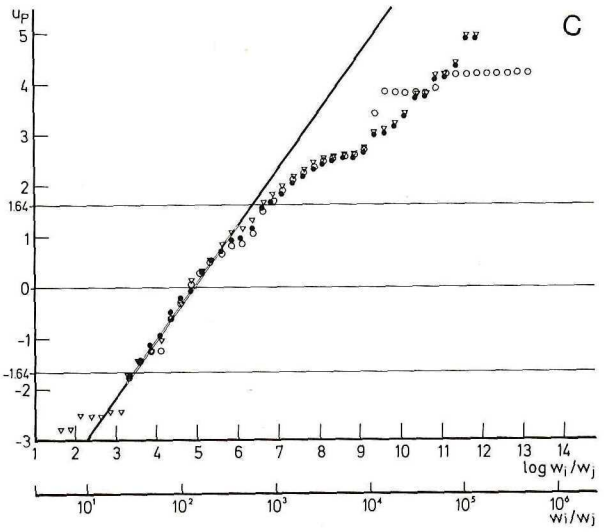
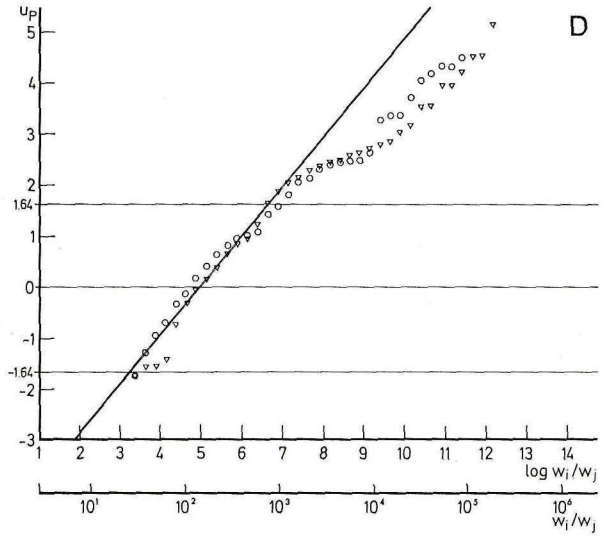


Fig. 4. A log-normal distribution fitted to cod prey size data: cumulative frequencies of $\log(w_i/w_j)$.

A: The entire material. The regression lines indicating bimodality are eye fitted.

B: The entire material (as in A) partitioned into records of cumaceans (triangles) and other food items (circles), with the estimated regression lines.





w_j).

C: The effect of dividing numbers of cumaceans by 100 (triangles). For comparison: records of cumaceans omitted (black dots, cf. B), and day hauls only ,cumaceans included (open circles). The regression line is eye fitted.

ems

D: The distributions for $w_i > 400$ g (open circles) and $w_i \leq 400$ g (triangles), cumaceans omitted. The regression line is that estimated for all sizes of cod, cumaceans omitted as in B.

quite acceptable considering that the "drooping tail" of the "other food items" regression curve concerns only 2 pCt of the distribution.

There are two obvious ways in which an overabundance of cumaceans can be brought about:–

1. Cumaceans are actually far more abundant than predicted by the model, eq. 3, for animals of that size.
2. Cumaceans occur in schools or patches of sufficient density for a large predator to consume several at a time.

Hypothesis 1 can be tested by comparing the predicted and observed numbers of cumaceans per cod stomach as a function of cod size. Cumaceans were all about the same size, $\overline{w_c} = .0142$g. Putting $w_j = \text{const.} = \overline{w_c}$ in eq. 5 gives

$$p\ (\log w_i) = \frac{1}{s\sqrt{2\pi}} \exp\left(-\frac{(\log w_i - \overline{\log w_i})^2}{2\,s^2}\right). \qquad (7)$$

The theoretical mean and variance can be substituted by the estimates for "other food items", see Fig. 4B. They are, mean $= \overline{\log(w_i/w_j)} = 4.974$, stand. dev. $= 1.04$. For $w_j = \overline{w_c}$ this gives $\log w_i = 4.974 + \log .0142 = .720$. Thus, the size of cod expected to eat cumaceans most frequently* is $w_i = \exp .720 = 2.05$g. Cod bigger than that should have fewer cumaceans in the stomach when cumaceans are considered all one size.

In the material from the Baltic in which cumaceans abound there are records of 1031 cumaceans distributed over 86 cod stomachs. Re-distributing these cumaceans according to eq. 7 results in the Figures of Table 4, which see.

The difference between observation and prediction is clear enough to warrant the conclusion that extraordinary abundance of cumaceans did not cause the bimodality of the distribution illustrated in Fig. 4A.

If the availability at night-time of small cumaceans as a source of food for large cod is due to a gregarious habit of cumaceans (several being perceived by the cod as one food item, hypothesis 2, above), the bimodality should be removed by dividing the numbers of cumaceans in each stomach by a constant or by an appropriate function of cod size.

The difference of the means of the two distributions illustrated in Fig. 4B ("cumaceans" and "other food items") is 9.48–4.97 = 4.51. The ratio of the medians of the two distributions of w_i/w_j is $\exp 4.51 \approx 10^2$. On preparing Fig. 4C (the triangles)

Table 4. Observed and predicted numbers of cumaceans per cod stomach as a function of cod size, assuming an extraordinary abundance of cumaceans. Material from the Baltic.

Cod weight group w_i	No. of cod	No. of cumaceans per stomach	
		Observed	Predicted
179	41	182	972
346	26	244	52.3
531	14	245	6.30
953	5	360	.267
Total.....	86	1031	1031

* Strictly: the size of cod eating equal numbers of food items bigger and smaller than cumaceans.

all numbers of cumaceans were divided by 100. Otherwise, everything is as in Fig. 4A. Black dots are the "other food items" of Fig. 4B. Open circles are day-hauls including cumaceans in their real numbers. The three data sets are notably like each other. The material does not invite more sophisticated delving into the mechanics at work. For instance, dividing cumacean numbers by 100 or by $w_i^{2/3}$ produces practically the same cumulative frequency curve. The latter corresponds to a mean divisor of 83, making "the number of cumaceans perceived as one" an increasing function of cod size (one may think of it as the area of the cod's open mouth).

The deviation from a log-normal distribution not yet accounted for (cf. Fig. 4C) is not serious as it concerns only the upper two per cent of the frequency distribution. It is caused by the occurrence, often one at a time, of some extremely small animals, perhaps consumed inadvertently. The slight indication of a remaining bimodality gives a suspicion that they were the stomach contents of the cod's own prey.

There remains to be investigated the hypothesis underlying the model that the frequency distribution of log (w_i/w_j) is independent of w_i, the predator size.

For that purpose all records excluding cumaceans were sorted into 234 records of $w_i \leq 400$g and 163 records of $w_i > 400$g. Inspection of the cumulative distributions in Fig. 4D shows that the hypothesis cannot be rejected by means of the present material.

Including outlying observations but omitting records of cumaceans the least squares parameter estimates for all sizes of cod are:–

	Mean $\left(\overline{\log \frac{w_i}{w_j}}\right)$	Variance
Actual environment	7.028	3.126
Simulated environment	5.114	1.089

The median of the predator/prey size relationship as obtained by antilogging the means are:–

Actual environment	1100:1
Simulated environment	160:1

The estimates for the actual environment are based on data in Table 3 for the distribution by numbers of prey animals in the stomachs. As in the case of dabs (preceding paragraph) the predator gets prey considerably smaller than preferred because of the greater abundance of small food items.

As compared with dabs, cod apparently prefer prey bigger by one size order, cf. Fig. 5. Remember that for dabs the variance is overestimated due to the way the data was presented.

CONCLUSION.

The model advanced is a fair description of the prey size preference when prey animals are distributed in the way most of them are (on aggregation in bottom animals, see Ursin 1960, p. 144). Individual prey species, however, may break the general pattern.

In an analytical approach the same model can be developed by assuming that on encounter with a prospective prey the predator evaluates the size of the prey relative to its own size and that the evaluation is symmetrical on a percentage scale so that,

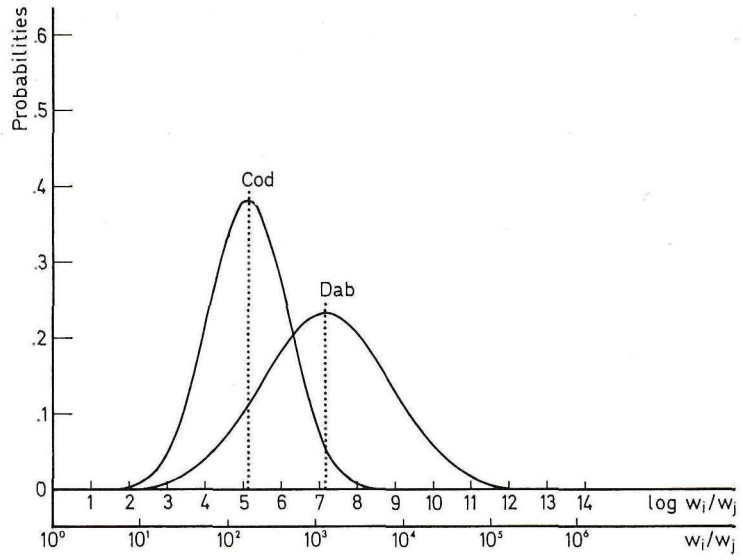


Fig. 5. Prey size selection in the simulated environment in which prey abundance is independent of prey size. Dab from Fig. 3.

e.g., "half as big" and "twice as big" differ by the same amount from "just fine". Thanks are due to Mr. K. P. Andersen who originally suggested to me the likelihood of a log-normal distribution of the predator-prey size ratio.

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